«abstract»
Entity
Models.java

- id : int

Entity(id: int)
+ getId() : int
+ *code() : String*
+ toString() : String
clean(s: String) : String

«class»
Payment
Models.java · extends Entity

- date : LocalDate
- amount : double
- method : String
- recordedBy : String
- payer : String

~ Payment(id: int, date: LocalDate, amount: double, method: String, recordedBy: String, payer: String)
+ code() : String
+ getDate() : LocalDate
+ setDate(date: LocalDate) : void
+ getAmount() : double
+ setAmount(amount: double) : void
+ getMethod() : String
+ setMethod(method: String) : void
+ getRecordedBy() : String
+ setRecordedBy(recordedBy: String) : void
+ getPayer() : String
+ setPayer(payer: String) : void

«class»
Data
Data.java

- users : ArrayList<User> = new ArrayList<>()
- clients : ArrayList<Client> = new ArrayList<>()
- cases : ArrayList<Case> = new ArrayList<>()
- appointments : ArrayList<Appointment> = new ArrayList<>()
- invoices : ArrayList<Invoice> = new ArrayList<>()
- documents : ArrayList<Doc> = new ArrayList<>()
- log : ArrayList<String[]> = new ArrayList<>()
- currentUser : User
- seeded : boolean
- loginError : String = ""
- nUser : int = 1
- nClient : int = 1
- nCase : int = 1
- nAppt : int = 1
- nInv : int = 1
- nPay : int = 1
- nDoc : int = 1
- userById : HashMap<Integer, User> = new HashMap<>()
- clientById : HashMap<Integer, Client> = new HashMap<>()
- caseById : HashMap<Integer, Case> = new HashMap<>()
- invoiceById : HashMap<Integer, Invoice> = new HashMap<>()

~ getUsers() : List<User>
~ getClients() : List<Client>
~ getCases() : List<Case>
~ getAppointments() : List<Appointment>
~ getDocuments() : List<Doc>
~ getLog() : List<String[]>
~ getCurrentUser() : User
~ setCurrentUser(u: User) : void
~ isSeeded() : boolean
~ removeUser(u: User) : void
~ removeClient(k: Client) : void
~ removeAppointment(a: Appointment) : void
~ removeInvoice(i: Invoice) : void
~ removeDocument(d: Doc) : void
~ removeCase(c: Case) : void
~ dataDir() : File
~ dataFile(name: String) : File
~ filesDir() : File
~ init() : void
~ clearAll() : void
~ newUser(username: String, password: String, fullName: String, email: String, phone: String, role: String) : User
~ newClient(name: String, email: String, phone: String) : Client
~ newCase(title: String, client: Client, lawyer: User, type: String, fee: double, status: String) : Case
~ newAppointment(event: String, c: Case, d: LocalDate, t: LocalTime, place: String, status: String) : Appointment
~ newInvoice(c: Case, description: String, issued: LocalDate, due: LocalDate, amount: double) : Invoice
~ newPayment(inv: Invoice, date: LocalDate, amount: double, method: String, recordedBy: String, payer: String) : Payment
~ newDoc(name: String, type: String, c: Case, location: String, status: String, addedBy: String) : Doc
~ seed() : void
~ findUser(username: String) : User
~ getLoginError() : String
~ login(username: String, password: String) : User
- systemLog(what: String) : void
~ usersByRole(role: String) : ArrayList<User>
~ visibleCases() : ArrayList<Case>
~ visibleAppointments() : ArrayList<Appointment>
~ visibleInvoices() : ArrayList<Invoice>
~ visibleDocuments() : ArrayList<Doc>
~ nextHearing(c: Case) : Appointment
~ invoicesOf(c: Case) : ArrayList<Invoice>
~ findConflict(lawyer: User, d: LocalDate, t: LocalTime, ignore: Appointment) : Appointment
~ verifyAdminPassword(password: String) : boolean
~ countAdmins() : int
~ commit(what: String) : void
- safe(s: String) : String
- line(fields: Object...) : String
- split(text: String) : String[]
- toDate(s: String) : LocalDate
- writeFile(name: String, columns: String, lines: ArrayList<String>) : void
- readFile(name: String) : ArrayList<String[]>
~ copyFile(from: File, to: File) : void
~ save() : boolean
- saveUsers() : void
- saveClients() : void
- saveCases() : void
- saveAppointments() : void
- saveInvoicesAndPayments() : void
- saveDocuments() : void
- saveLog() : void
~ load() : boolean
- loadUsers() : void
- loadClients() : void
- loadCases() : void
- loadAppointments() : void
- loadInvoices() : void
- loadPayments() : void
- loadDocuments() : void
- loadLog() : void

«abstract»
Person
Models.java · extends Entity

- name : String
- email : String
- phone : String

Person(id: int, name: String, email: String, phone: String)
+ getName() : String
+ setName(name: String) : void
+ getEmail() : String
+ setEmail(email: String) : void
+ getPhone() : String
+ setPhone(phone: String) : void
+ toString() : String

«class»
Doc
Models.java · extends CaseRecord

+ TYPES : String[] = {"Pleading", "Contract", …
+ STATUSES : String[] = {"Draft", "Filed", "Final…
- name : String
- type : String
- added : LocalDate
- location : String
- status : String
- addedBy : String

~ Doc(id: int, name: String, type: String, relatedCase: Case, added: LocalDate, location: String, status: String, addedBy: String)
+ code() : String
+ getName() : String
+ setName(name: String) : void
+ getType() : String
+ setType(type: String) : void
+ getAdded() : LocalDate
+ setAdded(added: LocalDate) : void
+ getLocation() : String
+ setLocation(location: String) : void
+ getStatus() : String
+ setStatus(status: String) : void
+ getAddedBy() : String
+ setAddedBy(addedBy: String) : void

«class»
LawFirmVol2
LawFirmVol2.java

+ main(args: String[]) : void

«class»
LoginPanel.ScalesIcon
LoginPanel.java (nested) · extends JComponent

~ ScalesIcon()
paintComponent(gr: Graphics) : void

«class»
Form
Page.java · extends JPanel

- grid : JPanel = new JPanel(new GridBagLay…
- error : JLabel = Ui.label(" ", Ui.BOLD, Ui…
- page : Page
- onSave : Saver
- row : int = 0
- col : int = 0
- first : JComponent

+ Form(page: Page, eyebrow: String, title: String, saveText: String, onSave: Saver)
+ submit() : void
+ section(title: String) : Form
+ add(label: String, input: JComponent, span: int) : JComponent
+ add(label: String, input: JComponent) : JComponent
+ focusFirst() : void

«class»
CellRenderer
TablePage.java · extends DefaultTableCellRenderer

~ TEXT : int = 0
~ BADGE : int = 1
~ ARROW : int = 2
- kind : int
- badge : String = ""

~ CellRenderer(kind: int)
+ getTableCellRendererComponent(t: JTable, v: Object, sel: boolean, focus: boolean, row: int, col: int) : Component
paintComponent(g0: Graphics) : void

«class»
ClientsPage
ClientsPage.java · extends TablePage

+ ClientsPage()
load() : void
- editClient() : void
- deleteClient() : void
- openClientForm(existing: Client) : void

«class»
DocumentsPage
DocumentsPage.java · extends TablePage

- typeBox : JComboBox<String>

+ DocumentsPage()
load() : void
onOpen(ref: Object) : void
- openFile() : void
- changeStatus() : void
- remove() : void
- addDocument() : void

«class»
PaymentsDialog
BillingPage.java · extends JDialog

- page : BillingPage
- invoice : Invoice
- model : DefaultTableModel
- table : JTable
- summary : JLabel = Ui.label(" ", Ui.BOLD, Ui…

+ PaymentsDialog(page: BillingPage, invoice: Invoice)
- reload() : void
- selectedPayment() : Payment
- showReceipt() : void
- editPayment() : void
- deletePayment() : void

«class»
Client
Models.java · extends Person

~ Client(id: int, name: String, email: String, phone: String)
+ code() : String

«class»
Appointment
Models.java · extends CaseRecord

+ CONFIRMED : String = "Confirmed"
+ TENTATIVE : String = "Tentative"
+ CANCELLED : String = "Cancelled"
- event : String
- date : LocalDate
- time : LocalTime
- location : String
- status : String

~ Appointment(id: int, event: String, relatedCase: Case, date: LocalDate, time: LocalTime, location: String, status: String)
+ code() : String
+ getEvent() : String
+ setEvent(event: String) : void
+ getDate() : LocalDate
+ setDate(date: LocalDate) : void
+ getTime() : LocalTime
+ setTime(time: LocalTime) : void
+ getLocation() : String
+ setLocation(location: String) : void
+ getStatus() : String
+ setStatus(status: String) : void
+ when() : LocalDateTime
+ isUpcoming() : boolean

«class»
Ui
Ui.java

+ baseRegular : Font
+ baseBold : Font
+ TITLE : Font
+ HEAD_F : Font
+ SUB : Font
+ BOLD : Font
+ PLAIN : Font
+ SMALL : Font
+ BIG : Font
+ TAG : Font
+ LOGO : Font
+ MONO : Font
+ NAVY : Color = new Color(15, 40, 70)
+ BLUE : Color = new Color(36, 84, 116)
+ BLUE_HI : Color = new Color(48, 104, 142)
+ GOLD : Color = new Color(150, 110, 40)
+ PAGE : Color = new Color(216, 220, 225)
+ SIDE : Color = new Color(235, 236, 238)
+ LIGHT : Color = new Color(245, 245, 245)
+ HEAD : Color = new Color(232, 234, 237)
+ ROW : Color = new Color(240, 243, 242)
+ LINE : Color = new Color(205, 208, 214)
+ GRAY : Color = new Color(110, 115, 125)
+ SOFT : Color = new Color(190, 200, 215)
+ AVATAR : Color = new Color(240, 228, 208)
+ TOPBAR : Color = new Color(243, 243, 243)
+ TEXT : Color = new Color(40, 45, 55)
+ RED : Color = new Color(179, 38, 30)
+ GREEN : Color = new Color(47, 107, 79)
~ stockiply : ImageIcon = new ImageIcon("images/law…
+ MOD : String = "Ctrl+"
+ MASK : int = java.awt.event.InputEvent…
- DATE_FMT : DateTimeFormatter = DateTimeFormatter.ofPatte…
- TIME_FMT : DateTimeFormatter = DateTimeFormatter.ofPatte…

+ loadFonts() : void
+ tipFor(buttonText: String) : String
+ tip(c: JComponent, text: String) : void
+ money(v: double) : String
+ peso(v: double) : Double
+ date(d: LocalDate) : String
+ time(t: LocalTime) : String
+ label(text: String, f: Font, c: Color) : JLabel
+ eyebrow(text: String) : JLabel
+ heading(eyebrow: String, title: String, east: JComponent) : JPanel
+ button(text: String, primary: boolean, listener: ActionListener) : Btn
+ primary(text: String) : Btn
+ secondary(text: String) : Btn
+ link(text: String, pageKey: String) : JLabel
+ inputBorder() : Border
+ field(text: String) : JTextField
+ password() : JPasswordField
+ area(rows: int) : JTextArea
+ scroll(c: JComponent) : JScrollPane
+ combo(items: String...) : JComboBox<String>
+ dateSpinner(d: LocalDate) : JSpinner
+ dateOf(s: JSpinner) : LocalDate
+ timeSpinner(t: LocalTime) : JSpinner
+ timeOf(s: JSpinner) : LocalTime
- toDate(t: LocalDateTime) : Date
+ statCard(caption: String, value: JLabel, numberColor: Color) : JPanel
+ section(title: String, right: JComponent, body: JComponent) : JPanel
+ warn(parent: Component, message: String) : void
+ info(parent: Component, message: String, title: String) : void
+ confirm(parent: Component, message: String) : boolean
+ ask(parent: Component, title: String, form: JComponent) : boolean
+ askAdminPassword(parent: Component, action: String) : boolean
+ showPrintable(parent: Component, title: String, text: String, fileName: String, width: int, height: int) : void
+ dialogForm() : JPanel
+ row(form: JPanel, label: String, input: JComponent) : void
+ names(list: List<User>) : String[]
+ clientNames(list: List<Client>) : String[]
+ caseLabels(list: List<Case>) : String[]
+ brand(height: int) : JPanel
+ avatar(initials: String, size: int) : JLabel

«class»
Receipt
BillingPage.java

- Receipt()
+ text(inv: Invoice, p: Payment) : String
+ show(parent: Component, inv: Invoice, p: Payment) : void

«class»
ActivityPage
AdminPages.java · extends TablePage

- stamp : DateTimeFormatter = DateTimeFormatter.ofPatte…

+ ActivityPage()
load() : void

«class»
User
Models.java · extends Person

+ ADMIN : String = "ADMIN"
+ LAWYER : String = "LAWYER"
+ MAX_ATTEMPTS : int = 3
+ ADMIN_LOCK_SECONDS : int = 30
- EMAIL : Pattern = Pattern.compile("^[^@\\s]…
- USERNAME : Pattern = Pattern.compile("^[A-Za-z…
- LAWYER_PERMS : String[] = { "VIEW_DASHBOARD", "VIEW…
- username : String
- role : String
- passwordHash : String
- failedAttempts : int = 0
- blocked : boolean = false
- lockedUntil : long = 0

~ User(id: int, username: String, fullName: String, email: String, phone: String, role: String)
+ code() : String
+ getUsername() : String
+ getRole() : String
+ setRole(role: String) : void
+ isActive() : boolean
+ getPasswordHash() : String
+ setPasswordHash(saved: String) : void
+ isValidEmail(s: String) : boolean
+ isValidUsername(s: String) : boolean
+ can(permission: String) : boolean
+ setPassword(password: String) : void
+ checkPassword(password: String) : boolean
- hashWithSalt(password: String, salt: byte[]) : String
+ isBlocked() : boolean
+ getFailedAttempts() : int
+ setFailedAttempts(n: int) : void
+ attemptsLeft() : int
+ block() : void
+ unblock() : void
+ lockSecondsLeft() : int
+ recordFailure() : boolean
+ recordSuccess() : void
+ shortName() : String
+ initials() : String
+ roleTitle() : String
+ statusText() : String

«class»
Shell
Shell.java · extends JPanel

~ current : Shell
~ statusLabel : JLabel = Ui.label(" ", Ui.SMALL, U…
~ NAV : String[][] = { {"Dashboard", "VIEW_DAS…
- root : CardLayout = new CardLayout()
- login : LoginPanel = new LoginPanel(this)
- visible : String = "login"
- appCard : JPanel
- pages : LinkedHashMap<String, Page>
- navItems : LinkedHashMap<String, JLabel>
- pageLayout : CardLayout
- pageHost : JPanel
- crumb : JLabel
- currentKey : String = "Dashboard"

+ Shell()
~ status(message: String) : void
- show(card: String) : void
+ loginAs(user: User) : void
~ logout() : void
+ navigate(key: String) : void
+ page(key: String) : Page
- createPage(key: String) : Page
- buildApp() : void
- buildSidebar(nav: JPanel, me: User) : JPanel
- buildTopBar(me: User) : JPanel
- changePassword() : void
- installShortcuts() : void
- shortcut(cmd: String) : void
- showShortcuts() : void
- navItem(text: String, key: String) : JLabel
- paintNav(l: JLabel, selected: boolean, hover: boolean) : void

«class»
HeaderRenderer
TablePage.java · extends DefaultTableCellRenderer

- owner : TablePage

~ HeaderRenderer(owner: TablePage)
+ getTableCellRendererComponent(t: JTable, v: Object, sel: boolean, focus: boolean, row: int, col: int) : Component

«class»
CasesPage
CasesPage.java · extends TablePage

- statusBox : JComboBox<String> = Ui.combo("All statuses", …
- typeBox : JComboBox<String>

+ CasesPage()
load() : void
onOpen(ref: Object) : void
~ addCase() : void
~ editCase() : void
- openCaseForm(existing: Case) : void
~ changeStatus() : void
~ removeCase() : void
~ showDetails(c: Case) : void

«class»
BillingPage
BillingPage.java · extends TablePage

- total : JLabel = new JLabel("0")
- received : JLabel = new JLabel("0")
- balance : JLabel = new JLabel("0")
- statusBox : JComboBox<String> = Ui.combo("All statuses", …

+ BillingPage()
load() : void
onRefreshed() : void
onOpen(ref: Object) : void
- showPayments() : void
- recordPayment() : void
- editInvoice() : void
- deleteInvoice() : void
- openInvoiceForm(existing: Invoice) : void
- printSummary() : void
- cut(s: String, n: int) : String

«class»
UsersPage
AdminPages.java · extends TablePage

- roleBox : JComboBox<String> = Ui.combo("All roles", Use…

+ UsersPage()
load() : void
- isSelf(u: User) : boolean
- lastAdmin(u: User) : boolean
- edit() : void
- reset() : void
- toggleBlock() : void
- delete() : void
- openUserForm(existing: User) : void
- nextId() : int

«class»
Case
Models.java · extends Entity

+ ACTIVE : String = "Active"
+ PENDING : String = "Pending"
+ HANDLED : String = "Handled"
+ STATUSES : String[] = {ACTIVE, PENDING, HANDLED}
+ TYPES : String[] = {"Civil", "Criminal", "Co…
- FIRM_NAME : String = "Specter Law Offices"
- FIRM_SHORT : String = "Specter"
- title : String
- client : Client
- lawyer : User
- type : String
- fee : double
- status : String
- opposing : String = ""
- court : String = ""
- filed : LocalDate
- opened : LocalDate
- description : String = ""

~ Case(id: int, title: String, client: Client, lawyer: User, type: String, fee: double, status: String)
+ getFirmName() : String
+ getFirmShort() : String
+ code() : String
+ getTitle() : String
+ setTitle(title: String) : void
+ getClient() : Client
+ getLawyer() : User
+ setLawyer(lawyer: User) : void
+ getType() : String
+ setType(type: String) : void
+ getFee() : double
+ setFee(fee: double) : void
+ getStatus() : String
+ setStatus(status: String) : void
+ getOpposing() : String
+ setOpposing(opposing: String) : void
+ getCourt() : String
+ setCourt(court: String) : void
+ getFiled() : LocalDate
+ setFiled(filed: LocalDate) : void
+ getOpened() : LocalDate
+ setOpened(opened: LocalDate) : void
+ getDescription() : String
+ setDescription(description: String) : void
+ isVisibleTo(u: User) : boolean
+ toString() : String

«class»
LoginPanel
LoginPanel.java · extends JPanel

- DARK : Color = new Color(16, 35, 60)
- RIGHT : Color = new Color(238, 238, 238)
- GOLD_ON_DARK : Color = new Color(184, 146, 70)
- TEXT_ON_DARK : Color = new Color(190, 200, 215)
- shell : Shell
- userField : JTextField
- passField : JPasswordField
- loginMessage : JLabel
- signIn : JButton
- showPassword : JCheckBox
- hiddenChar : char

+ LoginPanel(shell: Shell)
+ reset() : void
- buildLogin() : void
- buildLeft() : JPanel
- buildRight() : JPanel
- usernameFor(typed: String) : String
- doLogin() : void

«interface»
Saver
Page.java

+ *save() : String*

«abstract»
TablePage
TablePage.java · extends Page

~ PAGE_SIZE : int = 10
actions : JPanel = new JPanel(new FlowLayout…
filters : JPanel = new JPanel(new FlowLayout…
summary : JLabel = Ui.label(" ", Ui.BOLD, Ui…
search : JTextField
model : DefaultTableModel
table : JTable
columns : String[]
- exportName : String
- rows : ArrayList<Object[]> = new ArrayList<>()
- refs : ArrayList<Object> = new ArrayList<>()
- order : ArrayList<Integer> = new ArrayList<>()
- shown : ArrayList<Object> = new ArrayList<>()
- north : JPanel = new JPanel(new BorderLayo…
- pager : JPanel = new JPanel(new BorderLayo…
- pageNo : int = 0
~ sortCol : int = -1
~ sortAsc : boolean = true

TablePage(eyebrow: String, title: String, columns: String[], widths: int[], searchable: boolean)
setTop(c: JComponent) : void
load() : void
onRefreshed() : void
onOpen(ref: Object) : void
addRow(ref: Object, cells: Object...) : void
text(cell: Object) : String
+ refresh() : void
- compare(a: Object, b: Object) : int
- render() : void
- matches(row: Object[], q: String) : boolean
- buildPager(pages: int) : void
- clickButton(word: String) : boolean
+ shortcutNew() : void
+ shortcutEdit() : void
+ shortcutDelete() : void
+ shortcutFind() : void
+ shortcutPage(delta: int) : void
selected() : Object
pick(what: String) : Object
exportText() : void
- csv(s: String) : String
~ badgeColors(text: String) : Color[]
~ rowBg(sel: boolean, row: int) : Color

«class»
ReportsPage
ReportsPage.java · extends TablePage

- groupBox : JComboBox<String> = Ui.combo("By case type", …

+ ReportsPage()
load() : void

«abstract»
CaseRecord
Models.java · extends Entity

- relatedCase : Case

CaseRecord(id: int, relatedCase: Case)
+ getRelatedCase() : Case
+ setRelatedCase(relatedCase: Case) : void
+ getLawyer() : User

«class»
Invoice
Models.java · extends CaseRecord

+ METHODS : String[] = {"Cash", "Bank transfer",…
- description : String
- issued : LocalDate
- due : LocalDate
- amount : double
- payments : ArrayList<Payment> = new ArrayList<>()

~ Invoice(id: int, relatedCase: Case, description: String, issued: LocalDate, due: LocalDate, amount: double)
+ code() : String
+ getDescription() : String
+ setDescription(description: String) : void
+ getIssued() : LocalDate
+ setIssued(issued: LocalDate) : void
+ getDue() : LocalDate
+ setDue(due: LocalDate) : void
+ getAmount() : double
+ setAmount(amount: double) : void
+ getPayments() : List<Payment>
+ addPayment(p: Payment) : void
+ removePayment(p: Payment) : boolean
+ paid() : double
+ balance() : double
+ status() : String
+ checkPayment(amount: double, date: LocalDate, editing: Payment) : String
+ balanceAfter(payment: Payment) : double
~ round(v: double) : double

«class»
Btn
Ui.java · extends JButton

- primary : boolean

+ Btn(text: String, primary: boolean)
+ isPrimary() : boolean
paintComponent(g0: Graphics) : void

«abstract»
Page
Page.java · extends JPanel

- cards : CardLayout = new CardLayout()
- formHost : JPanel = new JPanel(new BorderLayo…
- activeForm : Form

Page()
setList(c: JComponent) : void
+ *refresh() : void*
+ onShow() : void
+ openForm(f: Form) : void
+ closeForm() : void
+ isFormOpen() : boolean
+ shortcutSave() : void
+ shortcutCancel() : void
+ shortcutNew() : void
+ shortcutEdit() : void
+ shortcutDelete() : void
+ shortcutFind() : void
+ shortcutPage(delta: int) : void
warn(message: String) : void
status(message: String) : void

«class»
RefreshListener
Page.java · implements ActionListener

- page : Page

~ RefreshListener(page: Page)
+ actionPerformed(e: ActionEvent) : void

«class»
DashboardPage
DashboardPage.java · extends Page

- values : JLabel[] = new JLabel[6]
- captions : String[]
- upcoming : DefaultTableModel
- tasks : JPanel = new JPanel()
- admin : boolean = Data.getCurrentUser().get…

+ DashboardPage()
+ refresh() : void
- set(i: int, v: int) : void
- buildTasks(mine: ArrayList<Case>) : void
- task(into: JPanel, title: String, detail: String) : int

«class»
SchedulePage
SchedulePage.java · extends TablePage

- bDay : JLabel = new JLabel("--", SwingCon…
- bKicker : JLabel = Ui.label(" ", Ui.SMALL, U…
- bTitle : JLabel = Ui.label(" ", Ui.SUB, Col…
- bPlace : JLabel = Ui.label(" ", Ui.SMALL, U…
- bWeek : JLabel = new JLabel("0", SwingCons…
- bFort : JLabel = new JLabel("0", SwingCons…
- lawyers : ArrayList<User> = Data.usersByRole(User.LAW…
- lawyerBox : JComboBox<String>
- periodBox : JComboBox<String> = Ui.combo("Upcoming", "Pas…

+ SchedulePage()
load() : void
onRefreshed() : void
onOpen(ref: Object) : void
- editSelected() : void
- cancelSelected() : void
- deleteSelected() : void
~ openAppointmentForm(existing: Appointment) : void
- showDetails(a: Appointment) : void
- buildBanner() : JPanel
- bannerStat(num: JLabel, caption: String) : JPanel
- updateBanner() : void

«class»
AssignmentsPage
AdminPages.java · extends TablePage

+ AssignmentsPage()
load() : void
- reassign() : void